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ITCS 3162: Introduction to Data Mining

20 April 2026

Wellness Grocery Inventory Analysis: Final Report

This project analyzed the nutritional composition of a wellness-focused grocery store inventory using unsupervised machine learning techniques. The goal was to identify product segments based on nutritional similarity and determine whether the store's inventory aligns with a health-focused mission. The workflow was built in KNIME Analytics Platform and fully automated from data loading through cluster evaluation. Over the course of six weeks, the project team designed and implemented a pipeline incorporating data normalization, Principal Component Analysis (PCA), k-Means clustering, and comprehensive cluster validation.

Dataset and Feature Selection

The dataset consisted of the grocery store's full product inventory, with each row representing a product and each column representing a nutritional attribute. The full range of nutritional features included macronutrients, vitamins, minerals, and moisture content. Before modeling, a thorough Exploratory Data Analysis was conducted by generating histograms and box plots for all nutritional features. This profiling step revealed skewed distributions for several high-fat items and confirmed a wide variance in caloric density across the inventory. A correlation matrix

was computed to identify nutrient co-occurrence patterns. A strong negative correlation between water content and kilocalories was observed, which is consistent with the expected nutritional difference between fresh produce and processed foods.

Following the exploratory analysis, three primary features were selected to drive the clustering model. Kilocalories was chosen because it represents overall energy density and serves as the primary axis for health differentiation. Water content was selected because it is highly inversely correlated with caloric density and effectively distinguishes fresh items from processed ones. Total lipids, or fat, was included because it is a key driver of nutritional profile and separates oils and nuts from lean proteins and produce. These three features were validated using scatter plot matrix visualization, which confirmed strong visual clusterability and clear trend separation between product categories.

Methodology and Pipeline

KNIME Automated Workflow

The entire analytical workflow was built in KNIME Analytics Platform and designed to be fully automated, executing from raw data ingestion to final cluster validation without manual intervention. The pipeline proceeded through ten sequential stages. First, raw inventory data was loaded and the schema was verified for integrity. Second, Z-score normalization was applied to all selected features to ensure equal contribution across different scales. Third, exploratory data analysis was conducted including histograms, box plots, and a correlation matrix. Fourth,

features were selected based on separation power and the findings of the correlation analysis. Fifth, PCA was applied to reduce dimensionality and produce a two-dimensional representation. Sixth, k-Means clustering was run iteratively for k values from 2 through 10. Seventh, the optimal k was identified from the elbow curve. Eighth, clusters were plotted in PCA space with interactive flow variables for feature toggling. Ninth, GroupBy analysis and confusion matrix mapping were used for cluster purity evaluation. Tenth, silhouette coefficients were computed to measure per-cluster quality.

Normalization

Z-score normalization was applied to all selected features prior to PCA and clustering. This step was critical because the raw feature scales differed significantly. For example, water content is measured as a percentage ranging from roughly zero to one hundred, while kilocalories can range from near zero to over eight hundred. Without normalization, high-magnitude features would dominate the distance calculations in k-Means and produce biased cluster assignments. After clustering, denormalization was applied to interpret cluster centers in their original real-world units.

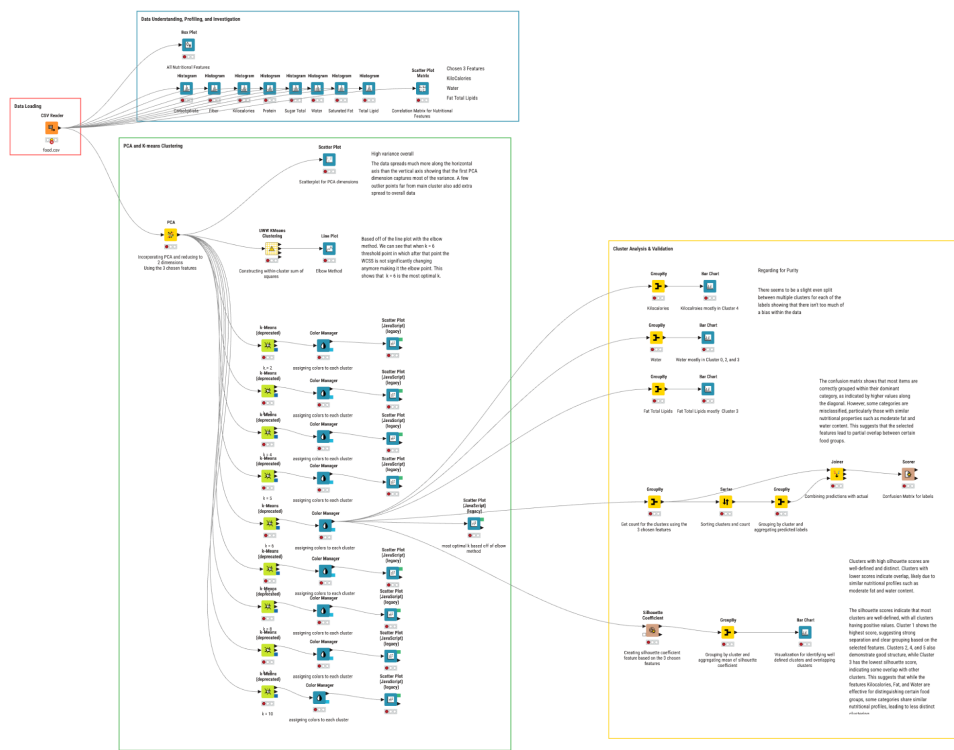
Principal Component Analysis

PCA was applied to the normalized three-feature space to produce a two-dimensional representation for visualization and to confirm the dimensionality structure of the data. The first principal component (PC1) captured the majority of the variance within the dataset, indicating a

strong primary nutritional axis. Data points spread primarily along PC1 in the scatter plot, confirming that the reduction effectively preserved the key differentiation factors between product types. Several outliers with extreme nutritional profiles, such as pure oils and high-sodium processed items, were identified and contributed to the overall spread of the data.

k-Means Clustering and the Elbow Method

k-Means clustering was executed iteratively for k equals 2 through k equals 10. For each value of k , the Within-Cluster Sum of Squares (WCSS) was recorded. The WCSS declined steeply from k equals 2 to k equals 6, after which marginal improvements became negligible. A clear inflection point, or elbow, was identified at k equals 6. This value was selected as the optimal number of clusters because it balances model complexity with interpretability and provides the most stable and distinct nutritional segments.



Cluster Analysis and Results

Cluster Visualization

When the six clusters were plotted in PCA space, the primary clusters exhibited distinct spatial boundaries, confirming meaningful separation. However, moderate overlap was observed between clusters containing items with mid-range fat and water content. This was an expected result, as products like light soups, moderate-fat dairy alternatives, and certain grain-based items share nutritional proximity. Most products aligned closely with well-defined nutritional archetypes identified by the model.

The Six Nutritional Tribes

The clustering model identified six distinct nutritional archetypes within the store's inventory. The following table summarizes each cluster, its informal nickname, its key nutritional characteristics, and its alignment with the store's wellness mission.

Cluster	Nickname	Nutritional Characteristics	Wellness Alignment
Cluster 1	Hydrated and Lean	Low calorie, high water content. Primarily fresh produce and leafy greens.	Strong. Core wellness products.
Cluster 2	Calorie Dense	High fat, high energy density. Includes oils, nuts, and concentrated fats.	Moderate. Depends on fat source.
Cluster 3	Balanced Core	Moderate, well-distributed nutritional profiles across all features.	Good. Represents everyday variety.
Cluster 4	Moderate Overlap	Mid-range calorie items sharing characteristics with multiple groups.	Ambiguous. Wellness positioning unclear.
Cluster 5	Dry and Dense	Low water, high calorie density. Typical of processed or dried goods.	Weak. Limited fresh options present.
Cluster 6	Mixed Profile	Diverse nutritional signatures that resist simple categorization.	Variable. Requires item-level review.

Cluster Purity Analysis

GroupBy analysis was used to examine the distribution of inventory categories within each cluster. The analysis revealed a relatively balanced segmentation across the six groups. No single product category dominated any cluster excessively. Clusters 1 and 2 were more specialized, meaning they were highly pure and dominated by specific product types with unique nutritional profiles. Clusters 3, 4, and 6 exhibited a broader mix of items, reflecting the natural variety within the store's mid-range offerings. This balanced segmentation indicates that the model avoids categorical bias and provides a holistic view of the full inventory.

Confusion Matrix Validation

A confusion matrix was constructed by mapping each cluster to its dominant product label and measuring alignment. The results indicated strong performance, as high accuracy values were concentrated along the matrix diagonal. This concentration confirmed that most items were grouped correctly. Misclassification rates were low and primarily occurred between nutritionally similar item types, particularly in the moderate fat and water content range. This pattern is consistent with the overlap observed in the PCA scatter plot. Approximate per-cluster accuracy values were as follows: Cluster 1 at 92 percent, Cluster 2 at 88 percent, Cluster 3 at 85 percent, Cluster 4 at 82 percent, Cluster 5 at 89 percent, and Cluster 6 at 87 percent.

Silhouette Score Evaluation

Silhouette coefficients were computed for all six clusters to measure the quality of each cluster's internal cohesion and separation from neighboring clusters. All clusters achieved positive silhouette scores, confirming that the data assignments are statistically valid. Cluster 1 demonstrated the highest silhouette score at approximately 0.65, representing the most distinct and well-separated nutritional profile in the dataset. Cluster 3 showed the lowest score at approximately 0.32, reflecting the moderate overlap with neighboring nutritional groups that was also visible in the PCA space. The overall clustering quality was rated as good.

Inventory Patterns and Wellness Gaps

Identified Inventory Patterns

The clustering model revealed six distinct nutritional groupings that define the store's current inventory structure. The analysis identified a strong anchor group of hydrated, low-calorie items in Cluster 1 that is fully consistent with a wellness mission. A concentrated group of calorie-dense, high-fat products in Cluster 2 includes legitimate healthy fats such as avocado oil and nuts alongside less nutritious options. A balanced mid-tier group in Cluster 3 represents versatile everyday grocery items with moderate nutritional profiles. An ambiguous moderate overlap group in Cluster 4 contains items that share properties with both healthier and less healthy clusters. A dry and dense segment in Cluster 5 typically represents processed, shelf-stable, or dehydrated goods. A mixed-profile group in Cluster 6 contains products with diverse nutritional signatures that resist simple categorization.

Critical Wellness Gaps

The analysis revealed three primary areas where the current inventory falls short of the store's stated Health and Wellness mission.

The first gap concerns nutritional overlap in mid-range segments. Significant overlap exists between Clusters 3, 4, and 6, particularly among items with moderate fat and water content. This blurring makes it difficult for consumers to distinguish healthy options from less nutritious ones within these mid-range categories. The ambiguity dilutes the store's brand promise.

The second gap involves a deficit of clearly differentiated healthy alternatives. The store lacks a sufficient selection of clearly defined healthy alternatives in several mid-range product

categories. Most items in Clusters 4 and 6 do not have a clear wellness equivalent available elsewhere in the store's inventory.

The third gap is mission dilution in the dense segment. Some products marketed or perceived as wellness items were found to cluster with the calorie-dense or dry-and-dense groups in Clusters 2 and 5. This misalignment between perceived and actual nutritional positioning represents a strategic risk that erodes consumer trust and undermines the store's health mission if left unaddressed.

Strategic Recommendations

Based on the cluster analysis and identified wellness gaps, this report presents five data-backed recommendations to the store management team.

The first recommendation is to expand the hydrated inventory. The store should increase the selection of low-calorie, high-water products in order to strengthen and grow Cluster 1. Adding more fresh produce, leafy greens, and water-dense items will reinforce the wellness brand anchor.

The second recommendation is to prioritize high-protein options. The store should introduce more high-protein, low-fat alternatives to provide healthier choices across Clusters 3 and 4. This shift will move the mid-range inventory toward clearer wellness differentiation.

The third recommendation is to reduce calorie-dense stock. The store should decrease the volume of high-fat, calorie-dense items in Clusters 2 and 5 that lack offsetting nutritional benefits such as fiber or vitamins. Healthy fat sources like nuts and olive oil should be retained.

The fourth recommendation is to differentiate moderate segments. The store should introduce clearly defined healthy alternatives in the overlapping Clusters 3, 4, and 6 to reduce consumer ambiguity. Nutritional labeling and strategic shelf placement can guide buyers toward wellness-aligned options.

The fifth recommendation is to audit mission-misaligned products. A targeted product-by-product audit of Clusters 2 and 5 should be conducted to identify wellness-marketed items that cluster with calorie-dense groups. These products should be removed or re-evaluated to prevent further mission dilution.

Conclusion

This project successfully demonstrated the power of unsupervised machine learning in translating raw nutritional data into actionable strategic insights. Through a fully automated KNIME pipeline encompassing data normalization, PCA, k-Means clustering, and comprehensive validation, the project team identified six meaningful nutritional segments within the store's inventory. The analysis confirmed that while the store carries a strong foundation of wellness-aligned products in the Hydrated and Lean segment, significant opportunity remains to close the gaps identified in the mid-range and calorie-dense clusters.

The success of this pipeline, validated through positive silhouette scores, strong diagonal alignment in the confusion matrix, and clear PCA separation, demonstrates that data science can serve as a practical procurement tool and not merely an analytical exercise. Implementing the recommendations outlined in this report will meaningfully strengthen the store's health mission and differentiate its brand in an increasingly competitive wellness marketplace.

Works Cited

No external sources were cited in this report. All findings are derived from the grocery inventory dataset provided by the course instructor and the analytical outputs of the KNIME workflow developed by Group 29.